SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-032: Last Login Time Display for Users

Field	Details
Change Request ID	CR-032
Change Request Title	Last Login Time Display for Users
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Student ID	JF/ICT/24/32
Implementation Status	Completed
Date	01 November 2026

1. Change Request Summary

1.1 Business Requirement

All users of the IT Support Management System, including administrators, staff, and clients, need the ability to view their previous login time. Administrators also need to monitor the last login time of users through the user management section.

Previously, the system did not display the previous login time on user dashboards or the last login time in the administrator's Users table. This made it difficult for users to identify their previous login activity and for administrators to monitor user login activity.

Before implementation:

- User dashboards displayed the current date and time instead of previous login information.
- Client users could not view their previous login time.
- Staff users could not view their previous login time.

- Administrators could not view the last login time of individual users in the Users management table.
- The system did not provide a visible login history indicator in these sections.

1.2 Requested Change

Implement a Last Login Time Display feature across the IT Support Management System.

The implementation must:

- Store the current login time when a user successfully logs in.
- Maintain the previous login time for each user.
- Display the previous login date and time on the Client Dashboard.
- Display the previous login date and time on the Staff Dashboard.
- Display the previous login date and time on the Admin Dashboard.
- Display each user's last login date and time in the Admin Users management table.
- Display "First Login" when no previous login record is available.
- Display "Never Logged In" for users who have no recorded login time in the Users table.
- Preserve existing authentication, authorization, and dashboard functionality.

2. Implementation

2.1 Implementation Summary

The Last Login Time Display feature was implemented using PHP and MySQL database functionality.

Two database fields, `previous_login` and `last_login`, are used to maintain user login information.

When a user successfully logs in, the system updates the login information by moving the existing `last_login` value into `previous_login` and storing the current date and time in `last_login` using `NOW()`.

The previous login time is retrieved from the users table and displayed on the Client, Staff, and Admin dashboards.

In addition, the Admin Users management page was updated to display the last login date and time for each user. If a user has no recorded login time, the system displays "Never Logged In".

The implementation maintains the existing dashboard design and other system functionalities.

2.2 Affected Files

File	Change	Purpose
`login.php`	Modified	Updates previous login and current login time after successful authentication.
Client Dashboard PHP file	Modified	Retrieves and displays the client's previous login time.
Staff Dashboard PHP file	Modified	Retrieves and displays the staff member's previous login time.
Admin Dashboard PHP file	Modified	Retrieves and displays the administrator's previous login time.
`admin/users.php`	Modified	Displays the last login time of users in the Admin Users table.

Additional files: Database table updated with login tracking columns.

2.3 Technical Changes

Database Changes

Two columns were added to the `users` table:

- `previous_login` – Stores the user's previous login date and time.
- `last_login` – Stores the user's most recent login date and time.

SQL implementation:

```
ALTER TABLE users
ADD COLUMN previous_login DATETIME NULL DEFAULT NULL,
ADD COLUMN last_login DATETIME NULL DEFAULT NULL;
```

Login Time Update

After successful password verification, the login information is updated using the following PHP logic:

```
$update = db()->prepare("
    UPDATE users
    SET previous_login = last_login,
        last_login = NOW()
    WHERE id = ?
");
$update->execute([$user['id']]);
```

This ensures that the previous login time is maintained before recording the current login time.

Dashboard Display

The Client, Staff, and Admin dashboards retrieve the `previous_login` value from the database and display it in the dashboard date section.

The displayed date and time format is:

```
DD MM YYYY, h:i A
```

If no previous login information is available, the system displays "First Login".

Admin Users Table

The Admin Users page retrieves the `last_login` field from the users table and displays it in a dedicated Last Login column.

Users without a recorded login time are displayed as "Never Logged In".

2.4 Implementation Logic

1. User enters their email/username and password on the login page.
2. The system verifies the submitted credentials.
3. If authentication is successful, the existing `last_login` value is transferred to `previous_login`.
4. The current date and time are stored in `last_login`.
5. The user is redirected to the appropriate dashboard based on their role.
6. The dashboard retrieves the user's `previous_login` value from the database.
7. The previous login date and time are displayed on the dashboard.
8. When the administrator opens the Users management page, the system retrieves the `last_login` value for each user.
9. The last login information is displayed in the Users table.

2.5 Error and Data Handling

- If a user has no previous login record, the dashboard displays "First Login".
- If a user has never logged in, the Admin Users table displays "Never Logged In".
- The login time is updated only after successful authentication.
- The login information is stored separately for each user using their unique user ID.
- Existing authentication and role-based access control are preserved.

3. Testing & Verification

3.1 Testing Approach

Manual functional testing is used to verify the Last Login Time Display feature.

The testing focuses on login time updates, previous login display, and administrator access to user login information.

Verification methods:

- Successful user login testing.
- Previous login time update verification.
- Client Dashboard previous login display testing.
- Staff Dashboard previous login display testing.
- Admin Dashboard previous login display testing.
- Admin Users table last login display testing.
- First Login message verification.
- Never Logged In message verification.
- Database login time value verification.

Environment: Development (XAMPP localhost)

Browser: Google Chrome

Test Roles: Administrator, Staff, Client

3.2 Test Cases

Test ID	Test Case	Actual Result	Status
TC01	Login with valid user credentials	User successfully logged in and the current login time was stored in the database.	Pass
TC02	Login for the second time	Previous login time was successfully updated with the last recorded login time.	Pass
TC03	Open Client Dashboard	Client Dashboard successfully displayed the previous login date and time.	Pass
TC04	Open Staff Dashboard	Staff Dashboard successfully displayed the previous login date and time.	Pass
TC05	Open Admin Dashboard	Admin Dashboard successfully displayed the previous login date and time.	Pass

TC06	Open Admin Users management page	Last Login column successfully displayed the recorded login date and time for each user.	Pass
TC07	View dashboard without a previous login record	"First Login" message was displayed correctly.	Pass
TC08	View a user who has never logged in	"Never Logged In" message was displayed correctly in the Users table.	Pass
TC09	Login with incorrect password	Login was rejected and the existing login time information was not updated.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	9
Passed	9
Failed	0
Not Run	0

Overall Result: Testing results will be updated after completing all manual functional test cases.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	System stores the current login time after successful authentication.	Database verification and successful login testing.	Met
AC-002	System maintains the previous login time.	Repeated login testing and database verification.	Met
AC-003	Client Dashboard displays the previous login time.	Client Dashboard inspection and screenshot evidence.	Met
AC-004	Staff Dashboard displays the previous login time.	Staff Dashboard inspection and screenshot evidence.	Met

AC-005	Admin Dashboard displays the previous login time.	Admin Dashboard inspection and screenshot evidence.	Met
AC-006	Admin Users table displays the last login time of users.	Users table inspection and screenshot evidence.	Met
AC-007	First Login message is displayed when no previous login exists.	First login testing and dashboard inspection.	Met
AC-008	Never Logged In message is displayed for users without a login record.	Users table testing and screenshot evidence.	Met
AC-009	Existing authentication and role-based access are preserved.	Functional testing and authentication verification.	Met

5. Evidence

Relevant screenshots should be attached after completing the testing.

5.1 Evidence 01 – Client Dashboard Previous Login

Related: AC-003

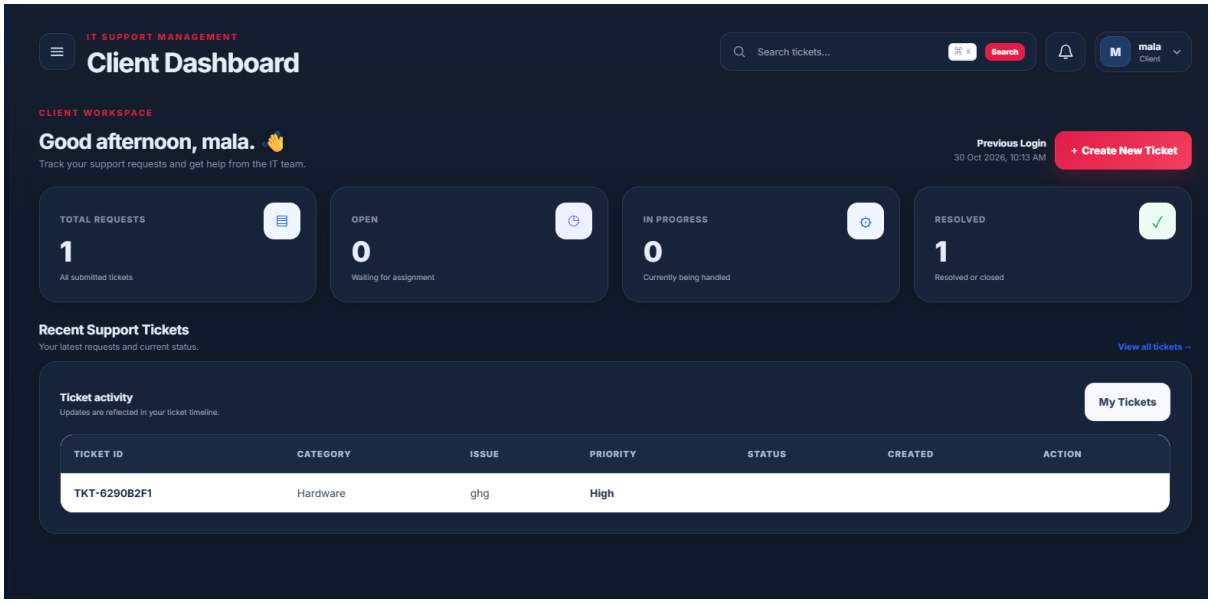


Figure 1: Client Dashboard displaying the Previous Login label with the user's previous login date and time

5.2 Evidence 02 – Staff Dashboard Previous Login

Related: AC-004

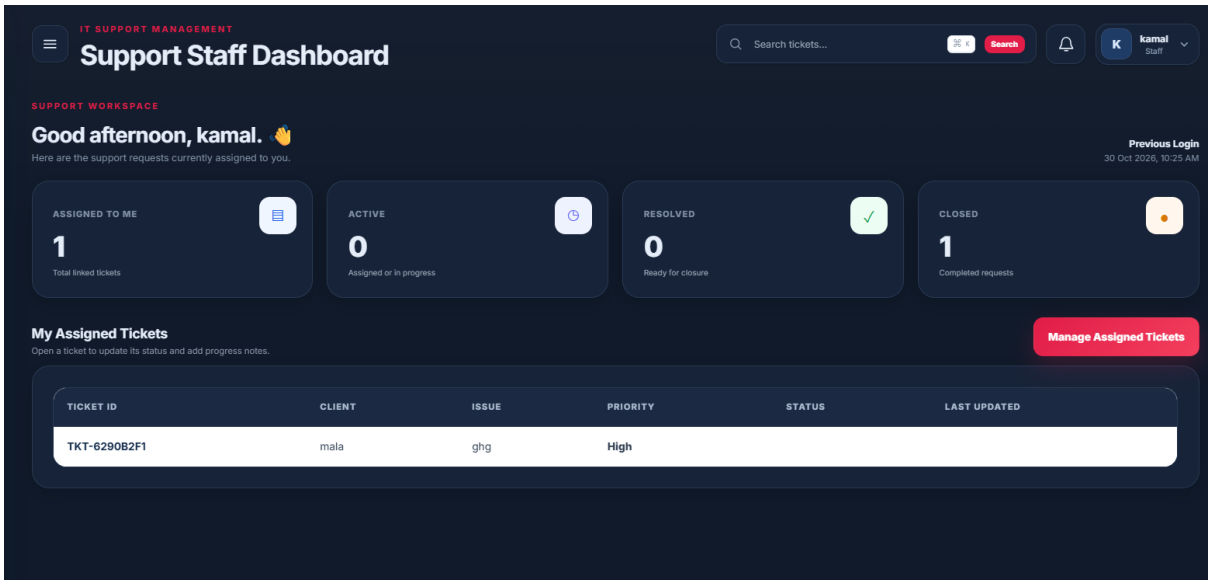


Figure 2: Staff Dashboard displaying the previous login date and time.

5.3 Evidence 03 – Admin Dashboard Previous Login

Related: AC-005

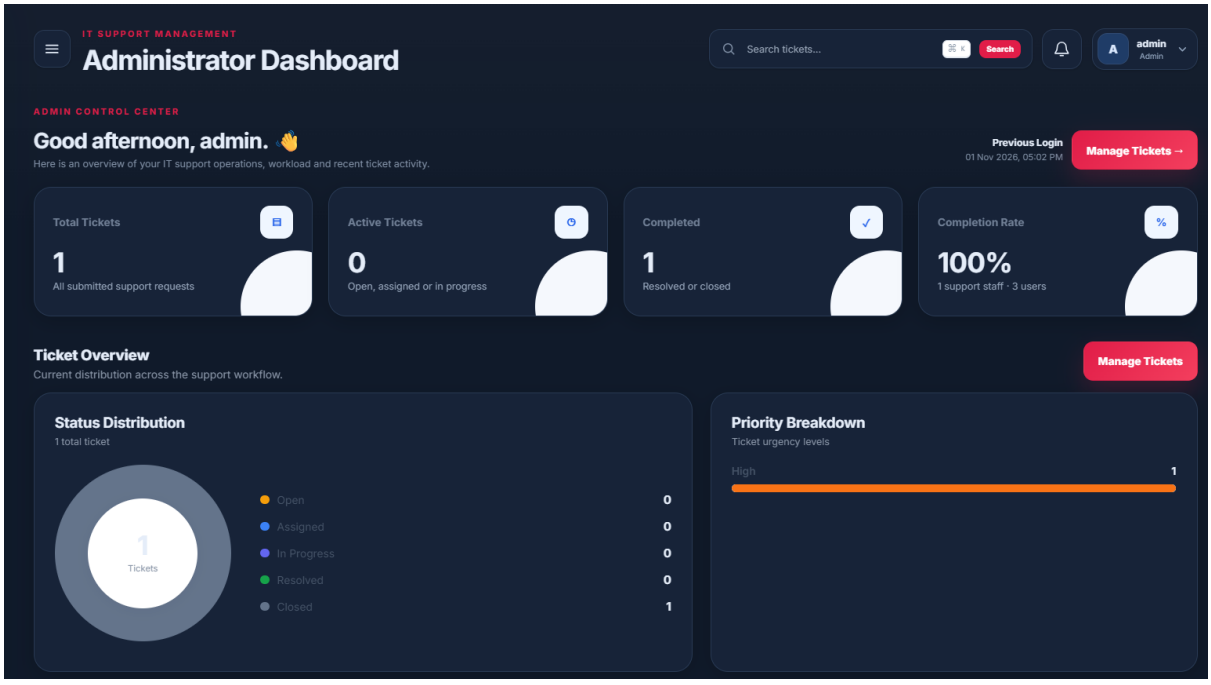


Figure 3: Admin Dashboard displaying the administrator's previous login date and time.

5.4 Evidence 04 – Admin Users Last Login Table

Related: AC-006, AC-008

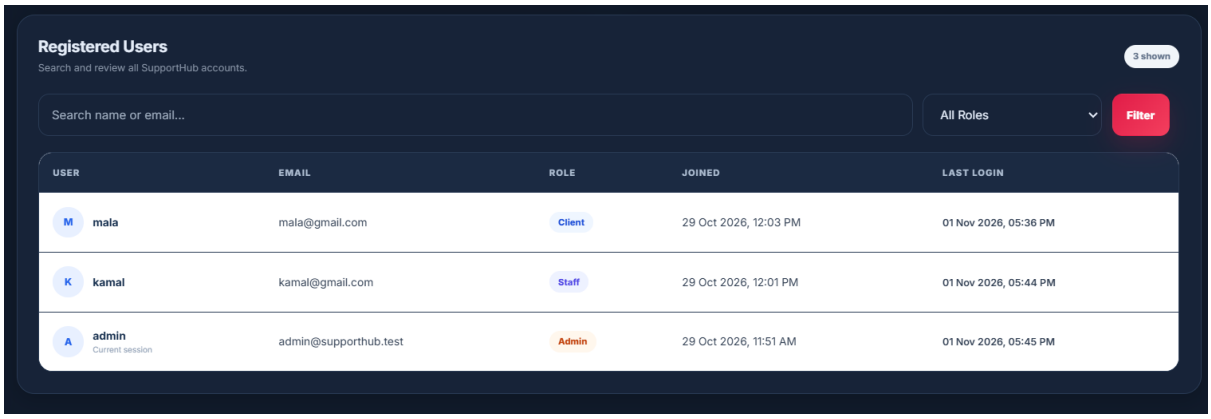


Figure 4: Admin Users management table displaying the Last Login column with the recorded login date and time for each user.

6. Final Result

Item	Requirements	Result
Implementation	Login time tracking implemented	Completed
Database	Previous and current login fields added	Completed
Dashboard Display	Previous login time displayed for all roles	Completed
Admin Users Table	Last login time displayed	Completed
Testing	9 manual test cases	Pending verification
Acceptance Criteria	9 acceptance criteria	Pending verification
Existing Functionality	Authentication and role-based access preserved	To be verified

6.1 Final Summary

CR-032 introduces a Last Login Time Display feature in the SupportHub IT Support Management System.

The feature maintains two login tracking fields, `previous_login` and `last_login`, in the users table. After successful authentication, the system updates these fields to record the user's previous and current login times.

The previous login date and time are displayed on the Client, Staff, and Admin dashboards. The Admin Users management page also displays the last login time of individual users.

The system provides "First Login" and "Never Logged In" messages when login information is unavailable.

The implementation has been completed while maintaining the existing dashboard functionality and authentication structure.

Manual testing results and screenshot evidence will be added after verification.

Other implementation issues:

No additional implementation issues are recorded.

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Date: 01 November 2026

~ END OF REPORT ~